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Thermal event detection and notification for vehicle batteries

Abstract

Methods, computing systems, and technology for thermal propagation monitoring and notification via air when vehicle is off are presented. For example, a computing system may be configured to obtain, while the vehicle is in an off state, battery data indicative of a state of a battery onboard the vehicle. The computing system may be configured to determine, based on the battery data, an occurrence of a thermal event onboard the vehicle while the vehicle is in the off state. The computing system may be configured to, in response to determining the occurrence of the thermal event, generate a message associated with the battery. The computing system may be configured to output, to one or more computing devices remote from the vehicle, the message associated with the battery.

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Background/Summary

FIELD

(1) The present disclosure relates generally to enhancing the ability of a vehicle to detect thermal events associated with its batteries. More particularly, the present disclosure relates to monitoring a battery onboard a vehicle while the vehicle is in an off state and sending notifications when a thermal event in the battery occurs.

BACKGROUND

(2) A vehicle such as an electric vehicle can include batteries for operating vehicle functions. For example, fully or hybrid electric vehicles can use batteries to help propel the electric vehicle. Batteries may be subject to thermal events such as thermal runaway. These events can damage a vehicle's battery infrastructure as well as other components of a vehicle.

SUMMARY

(3) Aspects and advantages of implementations of the present disclosure will be set forth in part in the following description, or may be learned from the description, or may be learned through practice of the implementations.

(4) One example aspect of the present disclosure is directed to a computing system of a vehicle. The computing system includes a control circuit configured to obtain, while the vehicle is in an off state, battery data indicative of a state of a battery onboard the vehicle. The control circuit is configured to determine, based on the battery data, an occurrence of a thermal event onboard the

vehicle while the vehicle is in the off state. The control circuit is configured to in response to determining the occurrence of the thermal event, generate a message associated with the battery. The control circuit is configured to output, to one or more computing devices remote from the vehicle, the message associated with the battery.

(5) In an example embodiment, the thermal event is a thermal runaway event.

(6) In an example embodiment, the control circuit is further configured to determine that the vehicle has transitioned from an on state to the off state.

(7) In an example embodiment, the vehicle is in a first battery monitoring mode when the vehicle is in the on state and a second battery monitoring mode when the vehicle is in the off state, the second battery monitoring mode being different from the first battery monitoring mode, and the control circuit is configured to output the message associated with the battery to the one or more computing devices remote from the vehicle when the vehicle is in the second battery monitoring mode.

(8) In an example embodiment, the vehicle is in a parked state.

(9) In an example embodiment, the message is indicative of the thermal event.

(10) In an example embodiment, the message is indicative of the state of the battery.

(11) In an example embodiment, the one or more computing devices remote from the vehicle include a plurality of user devices associated with users of the vehicle.

(12) In an example embodiment, the one or more computing devices remote from the vehicle are associated with a vehicle assistance service or an emergency provider.

(13) In an example embodiment, the one or more computing devices remote from the vehicle are included in a computing system onboard another vehicle.

(14) Another example aspect of the present disclosure is directed to a computer-implemented method. The computer-implemented method includes obtaining, while a vehicle is in an off state, battery data indicative of a state of a battery onboard the vehicle. The computer-implemented method includes determining, based on the battery data, an occurrence of a thermal event onboard the vehicle while the vehicle is in the off state. The computer-implemented method includes in response to determining the occurrence of the thermal event, generating a message associated with the battery. The computer-implemented method includes outputting, to one or more computing devices remote from the vehicle, the message associated with the battery.

(15) In an embodiment, the computer-implemented method includes determining that the vehicle has transitioned from an on state to the off state.

(16) In an embodiment, the computer-implemented method includes: predicting that the thermal event will occur; and outputting, to the one or more computing devices remote from the vehicle, an initial message indicating that the thermal event is predicted to occur.

(17) In an embodiment, the method includes outputting a signal to transition the vehicle to an on state or an awake state.

(18) In an embodiment, the one or more computing devices remote from the vehicle include at least one of: (i) a user device, (ii) a computing device associated with a vehicle assistance service, or (iii) a computing device associated with an emergency provider.

(19) In an embodiment, the one or more computing devices remote from the vehicle include a plurality of user devices associated with users of the vehicle.

(20) In an embodiment, the vehicle is being transported by another vehicle, and the one or more computing devices remote from the vehicle include a computing device onboard the other vehicle or a user device of an operator of the other vehicle.

(21) In an embodiment, the message is indicative of at least one of: (i) the thermal event, or (ii) the state of the battery.

(22) In an embodiment, the message is indicative of a location of the vehicle.

(23) Yet another example aspect of the present disclosure is directed to one or more non-transitory computer-readable media that store instructions that are executable by a control circuit to: obtain,

while a vehicle is in an off state, battery data indicative of a state of a battery onboard the vehicle; determine, based on the battery data, an occurrence of a thermal event onboard the vehicle while the vehicle is in the off state; in response to determining the occurrence of the thermal event, generate a message associated with the battery; and output, to one or more computing devices remote from the vehicle, the message associated with the battery.
(24) Other example aspects of the present disclosure are directed to other systems, methods, vehicles, apparatuses, tangible non-transitory computer-readable media, and devices for the technology described herein.
(25) These and other features, aspects, and advantages of various implementations will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate implementations of the present disclosure and, together with the description, serve to explain the related principles.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Detailed discussion of implementations directed to one of ordinary skill in the art are set forth in the specification, which makes reference to the appended figures, in which:
(2) FIG. **1** illustrates an example computing ecosystem according to an embodiment hereof.
(3) FIGS. **2**A-D illustrate diagrams of an example computing architecture for an onboard computing system of a vehicle according to an embodiment hereof.
(4) FIG. **3** illustrates an example vehicle interior with an example display according to an embodiment hereof.
(5) FIG. **4** illustrates a diagram of an example computing platform that is remote from a vehicle according to an embodiment hereof.
(6) FIG. **5** illustrates a diagram of an example user device according to an embodiment hereof.
(7) FIG. **6** illustrates a diagram of an example system for thermal event monitoring and notification according to an embodiment hereof.
(8) FIG. **7** illustrates a diagram of an example thermal event monitoring of a vehicle battery according to an embodiment hereof.
(9) FIG. **8** illustrates a diagram of an example system for thermal event monitoring and notification according to an embodiment hereof.
(10) FIG. **9** illustrates a flowchart diagram of an example method according to an embodiment hereof.
(11) FIG. **10** illustrates a diagram of an example computing ecosystem with computing components according to an embodiment hereof.
DETAILED DESCRIPTION
(12) Example aspects of the present disclosure are directed to monitoring a vehicle battery for the occurrence of a thermal event while the vehicle is in an off state and broadcasting a related message to remote computing devices. For example, a thermal event that may occur is thermal runaway, where the temperature of a battery cell increases uncontrollably. Other thermal events may include an internal short circuit, deformation, penetration, vibration, dendrites, articles, an external short circuit, over voltage, overcharge, over discharge, overcurrent, or external heating in a battery cell, as non-limiting examples. In some circumstances, these thermal events may lead to a thermal runaway of a battery.
(13) Thermal runaway, or another thermal event of the battery, can damage components of the vehicle as well as objects around the vehicle. For instance, thermal runaway can lead to thermal propagation when the thermal runaway in one battery cell spreads to other battery cells of the

battery, causing a fire in the vehicle.
(14) The technology of the present disclosure allows for monitoring a vehicle battery, determining that a thermal event, such as thermal runaway, has occurred while the vehicle is in the off state, and generating a message in response to determining that the thermal event occurred. For example, the vehicle may be parked with passengers inside or outside the vehicle. The battery can be monitored while the vehicle is parked and in an off state. A vehicle may be considered to be in an off state when the vehicle is not turned on, powered on, etc. A thermal event can be detected while the vehicle is parked and in an off state.
(15) A signal indicating that the thermal event occurred can be utilized to share information about the thermal event to computing devices outside the vehicle. For instance, in response to determining that the thermal event occurred, a message about the thermal event can be generated and sent to the vehicle passengers or users that are remote from the vehicle (e.g., a vehicle owner that is away from the vehicle). As a result, the vehicle passengers and users that are remote from the vehicle can take an appropriate action.
(16) The message may include information about the thermal event and the state of the vehicle battery, such as information indicating that thermal runaway is occurring in the vehicle battery. The message may also include information that identifies the location of the vehicle. The message can be output to computing devices that are outside of the vehicle. For example, the message can be sent over the air to user devices (e.g., the driver's phone and/or the passengers' phones) and computing devices (e.g., computers, phones, laptops, tablets, wearable devices, etc.) associated with emergency providers, vehicle assistance services, or another entity associated with the vehicle (e.g., maintenance provider, monitoring service). In some examples, the message can be sent over the air to computing devices onboard another vehicle where the message can be displayed on an instrument cluster in a vehicle that is nearby the vehicle experiencing the thermal event.
(17) The present disclosure provides a number of technical effects and computing improvements. For instance, the systems and methods of the present disclosure can increase the battery monitoring capabilities of the computing systems onboard the vehicle by continuously monitoring a battery of a vehicle. The computing systems can monitor the vehicle battery when the vehicle is in an on state and when the vehicle is in an off state. As such, the technology of the present disclosure improves the ability of the vehicle to determine that a thermal event has occurred at any time. This may be particularly useful when the vehicle is parked and in an off state and there are passengers in the vehicle who need to exit the vehicle if a thermal event occurs. This may also help avoid extensive damage caused by a thermal event by allowing possible mitigation from a user or remote service that is not within/onboard the vehicle.
(18) The technology of the present disclosure also improves the computing systems onboard the vehicle. For instance, signals generated while monitoring the battery can be utilized by the computing systems onboard the vehicle when the vehicle is in an on state and when the vehicle is in an off state. The signals can also be broadcast to other computing devices when the vehicle is on or when the vehicle is off. For example, the computing systems can use a signal generated while monitoring the battery when the vehicle is off to generate a message about the state of the battery and send the message to computing devices used by the users of the vehicle, emergency services, vehicle assistance services, and/or other vehicles. Accordingly, the vehicle computing system can avoid wasting its own computing resources by using and broadcasting the signals that are generated while the battery is monitored both when the vehicle is on and when the vehicle is off. In this way, the vehicle computing system can more efficiently utilize its computing resources.
(19) Reference now will be made in detail to embodiments, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the embodiments, not limitation of the present disclosure. In fact, it will be apparent to those skilled in the art that various modifications and variations may be made to the embodiments without departing from the scope or spirit of the present disclosure. For instance, features illustrated or described as part of one

embodiment may be used with another embodiment to yield a still further embodiment. Thus, it is intended that aspects of the present disclosure cover such modifications and variations.

(20) The technology of the present disclosure may include the collection of data associated with a user in the event that the user expressly authorizes such collection. Such authorization may be provided by the user via explicit user input to a user interface in response to a prompt that expressly requests such authorization. Collected data may be anonymized, pseudonymized, encrypted, noised, securely stored, or otherwise protected. A user may opt out of such data collection at any time.

(21) FIG. **1** illustrates an example computing ecosystem **100** according to an embodiment hereof. The ecosystem **100** may include a vehicle **105**, a remote computing platform **110** (also referred to herein as computing platform **110**), and a user device **115** associated with a user **120**. The user **120** may be a driver of the vehicle. In some implementations, the user **120** may be a passenger of the vehicle. In some implementations, the computing ecosystem **100** may include a third party (3P) computing platform **125**, as further described herein. The vehicle **105** may include a vehicle computing system **200** located onboard the vehicle **105**. The computing platform **110**, the user device **115**, the third-party computing platform **125**, and/or the vehicle computing system **200** may be configured to communicate with one another via one or more networks **130**.

(22) The systems/devices of ecosystem **100** may communicate using one or more application programming interfaces (APIs). This may include external facing APIs to communicate data from one system/device to another. The external facing APIs may allow the systems/devices to establish secure communication channels via secure access channels over the networks **130** through any number of methods, such as web-based forms, programmatic access via RESTful APIs, Simple Object Access Protocol (SOAP), remote procedure call (RPC), scripting access, etc.

(23) The computing platform **110** may include a computing system that is remote from the vehicle **105**. In an embodiment, the computing platform **110** may include a cloud-based server system. The computing platform **110** may be associated with (e.g., operated by) an entity. For example, the remote computing platform **110** may be associated with an OEM that is responsible for the make and model of the vehicle **105**. In another example, the remote computing platform **110** may be associated with a service entity contracted by the OEM to operate a cloud-based server system that provides computing services to the vehicle **105**.

(24) The computing platform **110** may include one or more back-end services for supporting the vehicle **105**. The services may include, for example, tele-assist services, navigation/routing services, performance monitoring services, etc. The computing platform **110** may host or otherwise include one or more APIs for communicating data to/from a vehicle computing system **200** of the vehicle **105** or the user device **115**.

(25) The computing platform **110** may include one or more computing devices. For instance, the computing platform **110** may include a control circuit and a non-transitory computer-readable medium (e.g., memory). The control circuit of the computing platform **110** may be configured to perform the various operations and functions described herein. Further description of the computing hardware and components of computing platform **110** is provided herein with reference to other figures.

(26) The user device **115** may include a computing device owned or otherwise accessible to the user **120**. For instance, the user device **115** may include a phone, laptop, tablet, wearable device (e.g., smart watch, smart glasses, headphones), personal digital assistant, gaming system, personal desktop devices, other hand-held devices, or other types of mobile or non-mobile user devices. As further described herein, the user device **115** may include one or more input components such as buttons, a touch screen, a joystick or other cursor control, a stylus, a microphone, a camera or other imaging device, a motion sensor, etc. The user device **115** may include one or more output components such as a display device (e.g., display screen), a speaker, etc. In an embodiment, the user device **115** may include a component such as, for example, a touchscreen, configured to

perform input and output functionality to receive user input and present information for the user **120**. The user device **115** may execute one or more instructions to run an instance of a software application and present user interfaces associated therewith, as further described herein. In an embodiment, the launch of a software application may initiate a user-network session with the computing platform **110**.

(27) The third-party computing platform **125** may include a computing system that is remote from the vehicle **105**, remote computing platform **110**, and user device **115**. In an embodiment, the third-party computing platform **125** may include a cloud-based server system. The term "third-party entity" may be used to refer to an entity that is different than the entity associated with the remote computing platform **110**. For example, as described herein, the remote computing platform **110** may be associated with an OEM that is responsible for the make and model of the vehicle **105**. The third-party computing platform **125** may be associated with a supplier of the OEM, a maintenance provider, a mapping service provider, an emergency provider, or other types of entities. In another example, the third-party computing platform **125** may be associated with an entity that owns, operates, manages, etc. a software application that is available to or downloaded on the vehicle computing system **200**.

(28) The third-party computing platform **125** may include one or more back-end services provided by a third-party entity. The third-party computing platform **125** may provide services that are accessible by the other systems and devices of the ecosystem **100**. The services may include, for example, mapping services, routing services, search engine functionality, maintenance services, entertainment services (e.g., music, video, images, gaming, graphics), emergency services (e.g., roadside assistance, 911 support), or other types of services. The third-party computing platform **125** may host or otherwise include one or more APIs for communicating data to/from the third-party computing platform **125** to other systems/devices of the ecosystem **100**.

(29) The networks **130** may be any type of network or combination of networks that allows for communication between devices. In some implementations, the networks **130** may include one or more of a local area network, wide area network, the Internet, secure network, cellular network, mesh network, peer-to-peer communication link or some combination thereof and may include any number of wired or wireless links. Communication over the networks **130** may be accomplished, for instance, via a network interface using any type of protocol, protection scheme, encoding, format, packaging, etc. In an embodiment, communication between the vehicle computing system **200** and the user device **115** may be facilitated by near field or short-range communication techniques (e.g., Bluetooth low energy protocol, radio frequency signaling, NFC protocol).

(30) The vehicle **105** may be a vehicle that is operable by the user **120**. In an embodiment, the vehicle **105** may be an automobile or another type of ground-based vehicle that is manually driven by the user **120**. For example, the vehicle **105** may be a Mercedes-Benz® car or van. In some implementations, the vehicle **105** may be an aerial vehicle (e.g., a personal airplane) or a water-based vehicle (e.g., a boat). The vehicle **105** may include operator-assistance functionality such as cruise control, advanced driver assistance systems, etc. In some implementations, the vehicle **105** may be a fully or semi-autonomous vehicle.

(31) The vehicle **105** may include a powertrain and one or more power sources. The powertrain may include a motor (e.g., an internal combustion engine, electric motor, or hybrid thereof), e-motor (e.g., electric motor), transmission (e.g., automatic, manual, continuously variable), driveshaft, axles, differential, e-components, gear, etc. The power sources may include one or more types of power sources. For example, the vehicle **105** may be a fully electric vehicle (EV) that is capable of operating a powertrain of the vehicle **105** (e.g., for propulsion) and the vehicle's onboard functions using electric batteries. In an embodiment, the vehicle **105** may use combustible fuel. In an embodiment, the vehicle **105** may include hybrid power sources such as, for example, a combination of combustible fuel and electricity.

(32) The vehicle **105** may include a vehicle interior. The vehicle interior may include the area

inside of the body of the vehicle **105** including, for example, a cabin for users of the vehicle **105**. The interior of the vehicle **105** may include seats for the users, a steering mechanism, accelerator interface, braking interface, etc. The interior of the vehicle **105** may include a display device such as a display screen associated with an infotainment system, as further described with respect to FIG. **3**.

(33) The vehicle **105** may include a vehicle exterior. The vehicle exterior may include the outer surface of the vehicle **105**. The vehicle exterior may include one or more lighting elements (e.g., headlights, brake lights, accent lights). The vehicle **105** may include one or more doors for accessing the vehicle interior by, for example, manipulating a door handle of the vehicle exterior. The vehicle **105** may include one or more windows, including a windshield, door windows, passenger windows, rear windows, sunroof, etc.

(34) The systems and components of the vehicle **105** may be configured to communicate via a communication channel. The communication channel may include one or more data buses (e.g., controller area network (CAN)), on-board diagnostics connector (e.g., OBD-II), or a combination of wired or wireless communication links. The onboard systems may send or receive data, messages, signals, etc. amongst one another via the communication channel.

(35) In an embodiment, the communication channel may include a direct connection, such as a connection provided via a dedicated wired communication interface, such as a RS-232 interface, a universal serial bus (USB) interface, or via a local computer bus, such as a peripheral component interconnect (PCI) bus. In an embodiment, the communication channel may be provided via a network. The network may be any type or form of network, such as a personal area network (PAN), a local-area network (LAN), Intranet, a metropolitan area network (MAN), a wide area network (WAN), or the Internet. The network may utilize different techniques and layers or stacks of protocols, including, e.g., the Ethernet protocol, the internet protocol suite (TCP/IP), the ATM (Asynchronous Transfer Mode) technique, the SONET (Synchronous Optical Networking) protocol, or the SDH (Synchronous Digital Hierarchy) protocol.

(36) In an embodiment, the systems/devices of the vehicle **105** may communicate via an intermediate storage device, or more generally an intermediate non-transitory computer-readable medium. For example, the non-transitory computer-readable medium, which may be external to the vehicle computing system **200**, may act as an external buffer or repository for storing information. In such an example, the vehicle computing system **200** may retrieve or otherwise receive the information from the non-transitory computer-readable medium.

(37) Certain routine and conventional components of vehicle **105** (e.g., an engine) are not illustrated and/or discussed herein for the purpose of brevity. One of ordinary skill in the art will understand the operation of conventional vehicle components in vehicle **105**.

(38) The vehicle **105** may include a vehicle computing system **200**. As described herein, the vehicle computing system **200** is onboard the vehicle **105**. For example, the computing devices and components of the vehicle computing system **200** may be housed, located, or otherwise included on or within the vehicle **105**. The vehicle computing system **200** may be configured to execute the computing functions and operations of the vehicle **105**.

(39) FIG. **2**A illustrates an overview of an operating system of the vehicle computing system **200**. The operating system may be a layered operating system. The vehicle computing system **200** may include a hardware layer **205** and a software layer **210**. The hardware and software layers **205**, **210** may include sub-layers. In some implementations, the operating system of the vehicle computing system **200** may include other layers (e.g., above, below, or in between those shown in FIG. **2**A). In an example, the hardware layer **205** and the software layer **210** can be standardized base layers of the vehicle's operating system.

(40) FIG. **2**B illustrates a diagram of the hardware layer **205** of the vehicle computing system **200**. In the layered operating system of the vehicle computing system **200**, the hardware layer **205** can reside between the physical computing hardware **215** onboard the vehicle **105** and the software

(e.g., of software layer **210**) that runs onboard the vehicle **105**.
(41) The hardware layer **205** may be an abstraction layer including computing code that allows for communication between the software and the computing hardware **215** in the vehicle computing system **200**. For example, the hardware layer **205** may include interfaces and calls that allow the vehicle computing system **200** to generate a hardware-dependent instruction to the computing hardware **215** (e.g., processors, memories, etc.) of the vehicle **105**.
(42) The hardware layer **205** may be configured to help coordinate the hardware resources. The architecture of the hardware layer **205** may be service oriented. The services may help provide the computing capabilities of the vehicle computing system **200**. For instance, the hardware layer **205** may include the domain computers **220** of the vehicle **105**, which may host various functionality of the vehicle **105** such as the vehicle's intelligent functionality. The specification of each domain computer may be tailored to the functions and the performance requirements where the services are abstracted to the domain computers. By way of example, this permits certain processing resources (e.g., graphical processing units) to support the functionality of a central in-vehicle infotainment computer for rendering graphics across one or more display devices for navigation, games, etc. or to support an intelligent automated driving computer to achieve certain industry assurances.
(43) The hardware layer **205** may be configured to include a connectivity module **225** for the vehicle computing system **200**. The connectivity module **225** may include code/instructions for interfacing with the communications hardware of the vehicle **105**. This can include, for example, interfacing with a communications controller, receiver, transceiver, transmitter, port, conductors, or other hardware for communicating data/information. The connectivity module **225** may allow the vehicle computing system **200** to communicate with other computing systems that are remote from the vehicle **105** including, for example, remote computing platform **110** (e.g., an OEM cloud platform).
(44) The architecture design of the hardware layer **205** may be configured for interfacing with the computing hardware **215** for one or more vehicle control units **230**. The vehicle control units **230** may be configured for controlling various functions of the vehicle **105**. This may include, for example, a central exterior and interior controller (CEIC), a charging controller, or other controllers as further described herein.
(45) The software layer **210** may be configured to provide software operations for executing various types of functionality and applications of the vehicle **105**. FIG. **2**C illustrates a diagram of the software layer **210** of the vehicle computing system **200**. The architecture of the software layer **210** may be service oriented and may be configured to provide software for various functions of the vehicle computing system **200**. To do so, the software layer **210** may include a plurality of sublayers **235**A-C. For instance, the software layer **210** may include a first sublayer **235**A including firmware (e.g., audio firmware) and a hypervisor, a second sublayer **235**B including operating system components (e.g., open-source components), and a third sublayer **235**C including middleware (e.g., for flexible integration with applications developed by an associated entity or third-party entity).
(46) The vehicle computing system **200** may include an application layer **240**. The application layer **240** may allow for integration with one or more software applications **245** that are downloadable or otherwise accessible by the vehicle **105**. The application layer **240** may be configured, for example, using container interfaces to integrate with applications developed by a variety of different entities.
(47) The layered operating system and the vehicle's onboard computing resources may allow the vehicle computing system **200** to collect and communicate data as well as operate the systems implemented onboard the vehicle **105**. FIG. **2**D illustrates a block diagram of example systems and data of the vehicle **105**.
(48) The vehicle **105** may include one or more sensor systems **305**. A sensor system may include or otherwise be in communication with a sensor of the vehicle **105** and a module for processing sensor data **310** associated with the sensor configured to acquire the sensor data **310**. This may

include sensor data **310** associated with the surrounding environment of the vehicle **105**, sensor data associated with the interior of the vehicle **105**, or sensor data associated with a particular vehicle function. The sensor data **310** may be indicative of conditions observed in the interior of the vehicle, exterior of the vehicle, or in the surrounding environment. For instance, the sensor data **310** may include image data, inside/outside temperature data, weather data, data indicative of a position of a user/object within the vehicle **105**, weight data, motion/gesture data, audio data, or other types of data. The sensors may include one or more: cameras (e.g., visible spectrum cameras, infrared cameras), motion sensors, audio sensors (e.g., microphones), weight sensors (e.g., for a vehicle a seat), temperature sensors, humidity sensors, Light Detection and Ranging (LIDAR) systems, Radio Detection and Ranging (RADAR) systems, or other types of sensors. The vehicle **105** may include other sensors configured to acquire data associated with the vehicle **105**. For example, the vehicle **105** may include inertial measurement units, wheel odometry devices, or other sensors.

(49) The vehicle **105** may include a positioning system **315**. The positioning system **315** may be configured to generate location data **320** (also referred to as position data) indicative of a location (also referred to as a position) of the vehicle **105**. For example, the positioning system **315** may determine location by using one or more of inertial sensors (e.g., inertial measurement units, etc.), a satellite positioning system, based on IP address, by using triangulation and/or proximity to network access points or other network components (e.g., cellular towers, WiFi access points, etc.), or other suitable techniques. The positioning system **315** may determine a current location of the vehicle **105**. The location may be expressed as a set of coordinates (e.g., latitude, longitude), an address, a semantic location (e.g., “at work”), etc.

(50) In an embodiment, the positioning system **315** may be configured to localize the vehicle **105** within its environment. For example, the vehicle **105** may access map data that provides detailed information about the surrounding environment of the vehicle **105**. The map data may provide information regarding: the identity and location of different roadways, road segments, buildings, or other items; the location and directions of traffic lanes (e.g., the location and direction of a parking lane, a turning lane, a bicycle lane, or other lanes within a particular roadway); traffic control data (e.g., the location, timing, or instructions of signage (e.g., stop signs, yield signs), traffic lights (e.g., stop lights), or other traffic signals or control devices/markings (e.g., cross walks)); or any other data. The positioning system **315** may localize the vehicle **105** within the environment (e.g., across multiple axes) based on the map data. For example, the positioning system **315** may process certain sensor data **310** (e.g., LIDAR data, camera data, etc.) to match it to a map of the surrounding environment to get an understanding of the vehicle's position within that environment. The determined position of the vehicle **105** may be used by various systems of the vehicle computing system **200** or another computing system (e.g., the remote computing platform **110**, the third-party computing platform **125**, the user device **115**).

(51) The vehicle **105** may include a communications unit **325** configured to allow the vehicle **105** (and its vehicle computing system **200**) to communicate with other computing devices. The vehicle computing system **200** may use the communications unit **325** to communicate with the remote computing platform **110** or one or more other remote computing devices over a network **130** (e.g., via one or more wireless signal connections). For example, the vehicle computing system **200** may utilize the communications unit **325** to receive platform data **330** from the computing platform **110**. This may include, for example, an over-the-air (OTA) software update for the operating system of the vehicle computing system **200**. Additionally, or alternatively, the vehicle computing system **200** may utilize the communications unit **325** to send vehicle data **335** to the computing platform **110**. The vehicle data **335** may include any data acquired onboard the vehicle including, for example, sensor data **310**, location data **320**, diagnostic data, user input data, data indicative of current software versions or currently running applications, occupancy data, data associated with the user **120** of the vehicle **105**, or other types of data obtained (e.g., acquired, accessed, generated,

downloaded, etc.) by the vehicle computing system **200**.
(52) In some implementations, the communications unit **325** may allow communication among one or more of the systems on-board the vehicle **105**.
(53) In an embodiment, the communications unit **325** may be configured to allow the vehicle **105** to communicate with or otherwise receive data from the user device **115** (shown in FIG. **1**). The communications unit **325** may utilize various communication technologies such as, for example, Bluetooth low energy protocol, radio frequency signaling, or other short range or near filed communication technologies. The communications unit **325** may include any suitable components for interfacing with one or more networks, including, for example, transmitters, receivers, ports, controllers, antennas, or other suitable components that may help facilitate communication.
(54) The vehicle **105** may include one or more human-machine interfaces (HMIs) **340**. The human-machine interfaces **340** may include a display device, as described herein. The display device (e.g., touchscreen) may be viewable by a user of the vehicle **105** (e.g., user **120**) that is located in the front of the vehicle **105** (e.g., driver's seat, front passenger seat). Additionally, or alternatively, a display device (e.g., rear unit) may be viewable by a user that is located in the rear of the vehicle **105** (e.g., back passenger seats). The human-machine interfaces **340** may present content via a user interface for display to a user **120**.
(55) The vehicle **105** may include a battery management system **360**. The battery management system **360** may monitor one or more batteries used by the vehicle **105**. The battery management system **360** may be configured to obtain battery data **365** related to the one or more batteries used by the vehicle **105**. The battery data **365** may include a charge level, voltage, capacity, leakage, gas levels, internal resistance, and temperature of a battery and the cells of a battery, as well as the state of the battery case, as non-limiting examples. The battery data **365** may be used by various systems of the vehicle computing system **200** or another computing system (e.g., the remote computing platform **110**, the third-party computing platform **125**, the user device **115**). For example, the vehicle computing system **200** may utilize the communications unit **325** to send the battery data **365** to the user device **115**.
(56) FIG. **3** illustrates an example vehicle interior **300** with a display device **345**. The display device **345** may be a component of the vehicle's head unit or infotainment system. Such a component may be referred to as a display device of the infotainment system or be considered as a device for implementing an embodiment that includes the use of an infotainment system. For illustrative and example purposes, such a component may be referred to herein as a head unit display device (e.g., positioned in a front/dashboard area of the vehicle interior), a rear unit display device (e.g., positioned in the back passenger area of the vehicle interior), an infotainment head unit or rear unit, or the like. The display device **345** may be located on, form a portion of, or function as a dashboard of the vehicle **105**. The display device **345** may include a display screen, CRT, LCD, plasma screen, touch screen, TV, projector, tablet, and/or other suitable display components.
(57) The display device may display a variety of content to the user **120** including information about the vehicle **105**, prompts for user input, etc. The display device may include a touchscreen through which the user **120** may provide user input to a user interface.
(58) For example, the display device **345** may include user interface rendered via a touch screen that presents various content. The content may include vehicle speed, mileage, fuel level, charge range, navigation/routing information, audio selections, streaming content (e.g., video/image content), internet search results, comfort settings (e.g., temperature, humidity, seat position, seat massage), or other vehicle data **335**. The display device **345** may render content to facilitate the receipt of user input. For instance, the user interface of the display device **345** may present one or more soft buttons with which a user **120** can interact to adjust various vehicle functions (e.g., navigation, audio/streaming content selection, temperature, seat position, seat massage, etc.). Additionally, or alternatively, the display device **345** may be associated with an audio input device

(e.g., microphone) for receiving audio input from the user **120**.
(59) Returning to FIG. 2D, the vehicle **105** may include a plurality of vehicle functions **350**A-C. A vehicle function **350**A-C may be a functionality that the vehicle **105** is configured to perform based on a detected input. The vehicle functions **350**A-C may include one or more: (i) vehicle comfort functions; (ii) vehicle staging functions; (iii) vehicle climate functions; (vi) vehicle navigation functions; (v) drive style functions; (v) vehicle parking functions; or (vi) vehicle entertainment functions. The user **120** may interact with a vehicle function **350**A-C through user input (e.g., to an adjustable input device, UI element) that specifies a setting of the vehicle function **350**A-C selected by the user.
(60) Each vehicle function may include a controller **355**A-C associated with that particular vehicle function **350**A-C. The controller **355**A-C for a particular vehicle function may include control circuitry configured to operate its associated vehicle function **350**A-C. For example, a controller may include circuitry configured to turn the seat heating function on, to turn the seat heating function off, set a particular temperature or temperature level, etc.
(61) In an embodiment, a controller **355**A-C for a particular vehicle function may include or otherwise be associated with a sensor that captures data indicative of the vehicle function being turned on or off, a setting of the vehicle function, etc. For example, a sensor may be an audio sensor or a motion sensor. The audio sensor may be a microphone configured to capture audio input from the user **120**. For example, the user **120** may provide a voice command to activate the radio function of the vehicle **105** and request a particular station. The motion sensor may be a visual sensor (e.g., camera), infrared, RADAR, etc. configured to capture a gesture input from the user **120**. For example, the user **120** may provide a hand gesture motion to adjust a temperature function of the vehicle **105** to lower the temperature of the vehicle interior.
(62) The controllers **355**A-C may be configured to send signals to another onboard system. The signals may encode data associated with a respective vehicle function. The encoded data may indicate, for example, a function setting, timing, etc. In an example, such data may be used to generate content for presentation via the display device **345** (e.g., showing a current setting). Additionally, or alternatively, such data can be included in vehicle data **335** and transmitted to the computing platform **110**.
(63) FIG. **4** illustrates a diagram of computing platform **110**, which is remote from a vehicle according to an embodiment hereof. As described herein, the computing platform **110** may include a cloud-based computing platform. The computing platform **110** may be implemented on one or more servers and include, or otherwise have access to, one or more databases. In an example, the computing platform **110** may be implemented using different servers based on geographic region.
(64) In some implementations, the computing platform **110** may include layered infrastructure that includes a plurality of layers. For instance, the computing platform **110** may include a cloud-based layer associated with functions such as security, automation, monitoring, and resource management. The computing platform **110** may include a cloud application platform layer associated with functions such as charging station functions, live traffic, vehicle functions, vehicle-sharing functions, etc. The computing platform **110** may include applications and services that are built on these layers.
(65) The computing platform **110** may be a modular connected service platform that includes a plurality of services that are available to the vehicle **105**. In an example, the computing platform **110** may include a container-based micro-services mesh platform. The services can be represented or implemented as systems within the computing platform **110**.
(66) In an example, the computing platform **110** may include a vehicle software system **405** that is configured to provide the vehicle **105** with one or more software updates **410**. The vehicle software system **405** can maintain a data structure (e.g., list, table) that indicates the current software or versions thereof downloaded to a particular vehicle. The vehicle software system **405** may also maintain a data structure indicating software packages or versions that are to be downloaded by the

particular vehicle. In some implementations, the vehicle software system **405** may maintain a data structure that indicates the computing hardware, charging hardware, or other hardware resources onboard a particular vehicle. These data structures can be organized by vehicle identifier (e.g., VIN) such that the computing platform **110** can perform a look-up function, based on the vehicle identifier, to determine the associated software (and updates) for a particular vehicle.
(67) When the vehicle **105** is connected to the computing platform **110** and is available to update its software, the vehicle **105** can request a software update from the computing platform. The computing platform **110** can provide the vehicle **105** one or more software updates **410** as over-the-air software updates via a network **130**.
(68) The computing platform **110** may include a remote assistance system **415**. The remote assistance system **415** may provide assistance to the vehicle **105**. This can include providing information to the vehicle **105** to assist with charging (e.g., charging locations recommendations), remotely controlling the vehicle (e.g., for AV assistance), roadside assistance (e.g., for collisions, flat tires), etc. The remote assistance system **415** may obtain assistance data **420** to provide its core functions. The assistance data **420** may include information that may be helpful for the remote assistance system **415** to assist the vehicle **105**. This may include information related to the vehicle's current state, an occupant's current state, the vehicle's location, the vehicle's route, charge/fuel level, incident data, etc. In some implementations, the assistance data **420** may include the vehicle data **335**.
(69) The remote assistance system **415** may transmit data or command signals to provide assistance to the vehicle **105**. This may include providing data indicative of relevant charging locations, remote control commands to move the vehicle, connect to an emergency provider, etc.
(70) The computing platform **110** may include a security system **425**. The security system **425** can be associated with one or more security-related functions for accessing the computing platform **110** or the vehicle **105**. For instance, the security system **425** can process security data **430** for identifying digital keys, data encryption, data decryption, etc. for accessing the services/systems of the computing platform **110**. Additionally, or alternatively, the security system **425** can store security data **430** associated with the vehicle **105**. A user **120** can request access to the vehicle **105** (e.g., via the user device **115**). In the event the request includes a digital key for the vehicle **105** as indicated in the security data **430**, the security system **425** can provide a signal to lock (or unlock) the vehicle **105**.
(71) The computing platform **110** may include a navigation system **435** that provides a back-end routing and navigation service for the vehicle **105**. The navigation system **435** may provide map data **440** to the vehicle **105**. The map data **440** may be utilized by the positioning system **315** of the vehicle **105** to determine a location of the vehicle **105**, a point of interest, etc. The navigation system **435** may also provide routes to destinations requested by the vehicle **105** (e.g., via user input to the vehicle's head unit). The routes can be provided as a portion of the map data **440** or as separate routing data. Data provided by the navigation system **435** can be presented as content on the display device **345** of the vehicle **105**.
(72) The computing platform **110** may include an entertainment system **445**. The entertainment system **445** may access one or more databases for entertainment data **450** for a user **120** of the vehicle **105**. In some implementations, the entertainment system **445** may access entertainment data **450** from another computing system (e.g., via an API) associated with a third-party service provider of entertainment content. The entertainment data **450** may include media content such as music, videos, gaming data, etc. The vehicle **105** may output the entertainment data **450** via one or more output devices of the vehicle **105** (e.g., display device, speaker, etc.).
(73) The computing platform **110** may include a user system **455**. The user system **455** may create, store, manage, or access user profile data **460**. The user profile data **460** may include a plurality of user profiles, each associated with a respective user **120**. A user profile may indicate various information about a respective user **120** including the user's preferences (e.g., for music, comfort

settings), frequented/past destinations, past routes, etc. The user profiles may be stored in a secure database. In some implementations, when a user **120** enters the vehicle **105**, the user's key (or user device) may provide a signal with a user or key identifier to the vehicle **105**. The vehicle **105** may transmit data indicative of the identifier (e.g., via its communications unit **325**) to the computing platform **110**. The computing platform **110** may look-up the user profile of the user **120** based on the identifier and transmit user profile data **460** to the vehicle computing system **200** of the vehicle **105**. The vehicle computing system **200** may utilize the user profile data **460** to implement preferences of the user **120**, present past destination locations, etc. The user profile data **460** may be updated based on information periodically provided by the vehicle **105**. In some implementations, the user profile data **460** may be provided to the user device **115**.

(74) FIG. **5** illustrates a diagram of example components of user device **115** according to an embodiment hereof. The user device **115** may include a display device **500** configured to render content via a user interface **505** for presentation to a user **120**. The display device **500** may include a display screen, CRT, LCD, plasma screen, touch screen, TV, projector, tablet, or other suitable display components. The user device **115** may include a software application **510** that is downloaded and runs on the user device **115**. In some implementations, the software application **510** may be associated with the vehicle **105** or an entity associated with the vehicle **105** (e.g., manufacturer, retailer, maintenance provider). In an example, the software application **510** may enable the user device **115** to communicate with the computing platform **110** and the services thereof.

(75) The technology of the present disclosure allows the vehicle computing system **200** to monitor a battery onboard the vehicle **105** while the vehicle is in an off state. More particularly, the vehicle computing system **200** may determine that a thermal event in the battery occurred while the vehicle is in the off state and use the network **130** to send notifications when the thermal event occurred to computing devices remote from the vehicle **105** (e.g., the user device **115**). As described herein, this technology can extend the ability of the vehicle **105** to detect a thermal event during both an on state and an off state of the vehicle **105** and to notify other computing devices of the thermal event.

(76) FIG. **6** illustrates a diagram of an example system **600** for thermal event monitoring and notification according to an embodiment hereof. The vehicle **105** may be configured to obtain the battery data **365** while the vehicle **105** is in an off state **620**. The off state **620** can include a state where the battery of the vehicle **105** is not powered, such as when the vehicle **105** is turned off and when the vehicle **105** is parked. The battery data **365** can be indicative of a state of a battery onboard the vehicle **105**. For instance, the battery management system **360** may monitor the battery temperature (e.g., via sensors of the vehicle **105** and the sensor systems **305**) while the vehicle **105** is in the off state **620** and the battery data **365** can indicate that the battery is overheating when the battery temperature is above a threshold.

(77) The battery data **365** can be obtained from the battery management system **360** of the vehicle computing system **200** while the vehicle **105** is in the off state **620**. For instance, the battery data **365** may be shared between the battery management system **360** and other components of the vehicle computing system **200** while the vehicle **105** is in the off state **620**. As a result, components of the vehicle computing system **200**, such as the communications unit **325**, can utilize the battery data **365** to send information about the battery to computing devices. Based on the battery data **365**, the vehicle **105** (e.g., the battery management system **360**) can determine that a thermal event **605** has occurred onboard the vehicle **105** while when the vehicle **105** is in the off state **620**.

(78) The vehicle **105** may be configured to determine that the vehicle **105** has transitioned from the on state **625** to the off state **620**. For instance, the vehicle **105** may determine that the vehicle **105** is turned off and the battery is not receiving power (e.g., via the battery management system **360**). When the vehicle **105** is in the on state **625**, the vehicle **105** may be in a first battery monitoring mode. When the vehicle **105** is in the off state **620**, the vehicle **105** may be in a second battery monitoring mode that differs from the first battery monitoring mode. The vehicle **105** may be

configured to output the message **610** to the computing devices **615** remote from the vehicle **105** when the vehicle **105** is in the second battery monitoring mode.

(79) In response to determining that the thermal event **605** has occurred, the vehicle **105** (e.g., the battery management system **360**) may generate a message **610** associated with the battery of the vehicle **105** that corresponds to the thermal event **605**. The message **610** may be indicative of at least one of the thermal event **635**, the state **640** of the battery, or the location **645** of the vehicle **105** (e.g., based on the location data **320**). For example, the message **610** may indicate that the state **640** of the battery is that the battery is overheating, that the thermal event **635** is thermal runaway, and that the location **645** of the vehicle **105** is at a set of coordinates (e.g., latitude, longitude).

(80) Thermal runaway can be detected by the battery management system **360** by measuring characteristics of a battery, such as voltage, temperature, and gas levels, which may be stored as the battery data **365**. For example, a voltage or current decrease or an impedance fluctuation may be detected by sensors in the battery (e.g., sensors of the vehicle **105** and the sensor systems **305**) and the battery management system **360** may determine that the voltage or current is anomalous and can cause thermal runaway. In another example, the sensors may measure the temperature of a battery or a battery cell and the battery management system **360** may determine that the temperature is at a level where thermal runaway may occur. For example, a battery may operate at a temperature between 77-100 degrees Fahrenheit or 25-37 degrees Celsius. A temperature measured by the sensors that is at the upper end of this range (e.g., 95 degrees Fahrenheit) or above this range may cause the battery management system **360** to determine that thermal runaway may occur. Thermal runaway can also be detected by sensors in a battery that detect gases and the battery management system **360** can use the information about the detected gases to determine that the gases may cause thermal runaway. For example, the sensors may detect any type of gas accumulation and the battery management system **360** may determine that thermal runaway may occur due to the gas accumulation. In another example, the sensors may measure the pressure of a battery or a battery cell and the battery management system **360** may determine that thermal runaway may occur when there is any detection by the sensors of battery pressure increase.

(81) In some implementations, a threshold (e.g., temperature, gas accumulation, pressure, etc.) for detecting a thermal event may depend on the state of the vehicle. For example, the temperature threshold for detecting a thermal event in an on state may be different than a temperature threshold for detecting a thermal event in an off state. In an on state, the temperature may range between 75-100 F, while the high voltage of the vehicle is active/the vehicle is on. The temperature threshold for detecting a thermal event may be higher for the on state than for an off state, while the high voltage of the vehicle is inactive/the vehicle is off. The battery management system **360** may switch its threshold monitoring depending on the state of the vehicle. For example, the battery management system **360** may monitor for a first threshold for the on state and a second threshold for the off state. The battery management system **360** may access a look-up table that indicates the respective thresholds when transition between its modes to implement the appropriate thresholds.

(82) The vehicle **105** (e.g., the communications unit **325**) may output the message **610** to one or more computing devices **615** remote from the vehicle **105**. The computing devices **615** remote from the vehicle **105** can include at least one of a vehicle assistance service **650**, an emergency provider **655**, the user device **115** associated with the user **120**, or a computing system onboard another vehicle **660**. For example, the message **610** may be sent over the air from the vehicle **105** to the user device **115** associated with the user **120** of the vehicle **105** and to the emergency provider **655** to provide assistance at the location of the vehicle **105**. The computing devices **615** remote from the vehicle **105** can also include a plurality of user devices associated with users of the vehicle. For instance, the message **610** may be sent over the air from the vehicle **105** (e.g., the communications unit **325**) to the user devices (e.g., computers, phones, laptops, tablets, wearable devices, etc.) associated with the driver and the passengers of the vehicle **105**. In some implementations, the user **120**, such as the owner of the vehicle **105**, can use a mobile application

associated with the vehicle **105** to identify the users and user devices to send the message **610** to when the thermal event **605** occurs. In other implementations, users in addition to the owner of the vehicle **105** can use a mobile application associated with the vehicle **105** to configure their user device **115** to receive the message **610** when the thermal event **605** occurs onboard the vehicle **105**.
(83) For example, the vehicle **105** may be parked overnight in a home garage of the vehicle owner when the vehicle **105** determines that the thermal event **605** occurred onboard the vehicle **105**. The message **610** can be generated and sent to the user device **115** of the owner and to the emergency provider **655** so that the owner can contact emergency services, etc.
(84) In another example, the vehicle **105** may be in a manufacturing plant or in a fleet of parked vehicles, where the occurrence of the thermal event **605** onboard the vehicle **105** can affect other vehicles that are close by if a fire occurs due to the thermal event **605**. The vehicle **105** can determine that the thermal event **605** occurred and the message **610** can be generated and sent to the user device **115** of the plant owner, plant employees, fleet owner, or another user, and to computing systems onboard the other vehicles in the manufacturing plant and the fleet, so that the other vehicles can be moved away from the vehicle **105** with the thermal event **605**.
(85) In some implementations, a signal may be output (e.g., by the vehicle computing system **200**) to transition the vehicle **105** to an on state **625** or an awake state. For example, the message **610** may be output to the display device **345**, which may be a component of the vehicle's head unit or infotainment system. The display device **345** may require power in order to display the message **610**. In this example, the signal can be output to transition the vehicle **105** to the on state **625** in order for the message **610** to be displayed on the display device **345**. In some examples, a sound, such as the vehicle **105** horn or an internal buzzer, can sound in addition to the message **610** to be displayed on the display device **345**. Transitioning the vehicle **105** to the on state **625** in order to display the message **610** on the display device **345** can allow users who are in the vehicle **105** while the vehicle **105** is in the off state **620** and parked to leave the vehicle **105** when the thermal event **605** occurs onboard the vehicle **105**. Transitioning the vehicle **105** to the on state **625** can also provide power to the vehicle **105** to allow the vehicle computing system **200** to take steps to mitigate the thermal event **605**, such as cooling the battery of the vehicle **105**.
(86) In some examples, the vehicle **105** (e.g., the battery management system **360**) may be configured to predict that the thermal event **605** will occur. The vehicle **105** (e.g., the communications unit **325**) may output the message **610** to the one or more computing devices **615** remote from the vehicle **105**, and the message **610** can be an initial message with a thermal event prediction **630** that indicates that the thermal event **605** is predicted to occur. For example, the battery management system **360** may monitor the battery temperature (e.g., via sensors of the vehicle **105** and the sensor systems **305**) while the vehicle **105** is in the off state **620** and the battery data **365** can indicate that the battery temperature is reaching a threshold level where the thermal event **605** is predicted to occur. In another example, sensors (e.g., the sensor system **305**) may detect that the battery of the vehicle **105** has been punctured and that the thermal event **605** is predicted to occur due to the damage to the battery. The message **610** can be output to the one or more computing devices **615** remote from the vehicle **105** and include the thermal event prediction **630** indicating that the thermal event **605** is predicted to occur, allowing for more time to take action before the thermal event **605** occurs.
(87) FIG. **7** illustrates a diagram of an example thermal event monitoring **700** of a vehicle battery according to an embodiment hereof. The thermal event **605** may be thermal runaway **710**. A battery **705** of the vehicle **105** can generate heat as the battery **705** is being used. Thermal runaway **710** may occur when the temperature of a battery cell of the battery **705** of the vehicle **105** increases uncontrollably. Thermal runaway **710** can be caused by a variety of factors, such as mechanical issues (e.g., a punctured battery cell), electrical issues (e.g., overcharging), or thermal issues (e.g., overheating). Thermal runaway **710** can further cause thermal propagation when the thermal runaway **710** in one battery cell of the battery **705** of the vehicle **105** spreads to other battery cells

of the battery **705**.
(88) While the vehicle **105** is in the off state **620**, the vehicle **105** (e.g., the battery management system **360**) can determine that the thermal event **605** of thermal runaway **710** occurred onboard the vehicle **105**. The vehicle **105** (e.g., the battery management system **360**) can generate the message **610** associated with the battery **705** of the vehicle **105** that corresponds to the thermal runaway **710** and output the message **610** to the computing devices **615** remote from the vehicle **105**. In this manner, others (e.g., the user **120**, the vehicle assistance service **650**, the emergency provider **655**, etc.) can be alerted and take an appropriate action. Notification at the time of the thermal runaway **710** can allow for a quick response before thermal propagation occurs.
(89) FIG. **8** illustrates a diagram of an example system **800** for thermal event monitoring and notification according to an embodiment hereof. In the example of FIG. **8**, the vehicle **105** may be transported by another vehicle **805**. The other vehicle **805** may be, for example, a vehicle with a trailer, a car carrier trailer, or a semi-trailer truck, that is transporting the vehicle **105**. While the vehicle **105** is in the off state **620** on the other vehicle **805**, the vehicle **105** (e.g., the battery management system **360**) may determine, based on the battery data **365** of the vehicle computing system **200**, that the thermal event **605** occurred onboard the vehicle **105**.
(90) The vehicle **105** (e.g., the battery management system **360**) can generate the message **610** associated with the battery of the vehicle **105** that corresponds to the thermal event **605** and output the message **610** to the computing devices **615** remote from the vehicle **105**, which can include a computing device onboard the other vehicle **805** or a user device **115** of an operator of the other vehicle **805**. For example, a car carrier trailer may be transporting vehicles with batteries, such as the vehicle **105**, and the operator of the car carrier trailer can receive the message **610** on their user device **115** when the thermal event **605** occurs onboard one of the vehicles being transported so that the operator can take an appropriate action. The user device **115** of the operator of the other vehicle **805** may have an application that identifies the vehicles that are being transported on the other vehicle **805**. For instance, the operator can use the user device **115** to scan a QR code or VIN of each vehicle being transported in order for the operator to receive a message **610** on the user device **115** when one of the vehicles being transported experiences a thermal event **605**. In another example, the message **610** may be output to a computing system, such as a display device, of the other vehicle **805** that is transporting the vehicles so that the operator can move the other vehicle **805** off the road or away from other vehicles when the thermal event **605** occurs onboard one of the vehicles being transported.
(91) FIG. **9** illustrates a flowchart diagram of an example method **900** for thermal event monitoring and notification according to an embodiment hereof. The method **900** may be performed by a computing system described with reference to the other figures. In an embodiment, the method **900** may be performed by the control circuit of the vehicle computing system **200** of FIG. **1**. One or more portions of the method **900** may be implemented as an algorithm on the hardware components of the devices described herein (e.g., as in FIGS. **1-8** and **10**, etc.). For example, the steps of method **900** may be implemented as operations/instructions that are executable by computing hardware.
(92) FIG. **9** illustrates elements performed in a particular order for purposes of illustration and discussion. Those of ordinary skill in the art, using the disclosures provided herein, will understand that the elements of any of the methods discussed herein may be adapted, rearranged, expanded, omitted, combined, or modified in various ways without deviating from the scope of the present disclosure. FIG. **9** is described with reference to elements/terms described with respect to other systems and figures for example illustrated purposes and is not meant to be limiting. One or more portions of method **900** may be performed additionally, or alternatively, by other systems. For example, method **900** may be performed by a control circuit of the computing platform **110**.
(93) In an embodiment, the method **900** may begin with or otherwise include an operation **905**, in which, the vehicle computing system **200** may determine that the vehicle **105** has transitioned from

an on state **625** to the off state **620**. For instance, a user **120** may park a vehicle **105** in the user's garage. As such, the vehicle **105** may be in a parked state. The user **120** may turn off the vehicle **105** such that the vehicle transitions from an on state **625** to an off state **620**. The vehicle computing system **200** may detect this transition by monitoring one or more components of the vehicle **105** and determining that power is not being supplied to such components.

(94) The method **900** in an embodiment may include an operation **910**, in which the vehicle computing system **200** may obtain, while the vehicle **105** is in an off state **620**, battery data **365** indicative of a state of a battery **705** onboard the vehicle **105**. For instance, the vehicle **105** may be in a first battery monitoring mode when the vehicle **105** is in the on state **625** and a second battery monitoring mode when the vehicle **105** is in the off state **620**, the second battery monitoring mode being different from the first battery monitoring mode. For example, the user **120** may turn off the vehicle **105** after parking the vehicle **105** in the user's garage, such that the vehicle **105** is in the off state **620**. The vehicle computing system **200** of the vehicle **105** may obtain battery data **365** indicating that the state of the battery **705** onboard the vehicle is such that the battery **705** is in a second battery monitoring mode because the vehicle **105** is in the off state **620**.

(95) The vehicle **105** may be configured differently when in the different modes, as described herein. For example, a message **610** associated with the battery **705** may be output to the one or more computing devices **615** remote from the vehicle **105** when the vehicle **105** is in the second battery monitoring mode. This may be different from an on state **625**, in which the message **610** may alternatively, or additionally, be transmitted to a head unit **347** of the vehicle **105**.

(96) The method **900** in an embodiment may include an operation **915**, in which the vehicle computing system **200** may predict that a thermal event **605** will occur. The thermal event **605** may be a thermal runaway **710** event. For instance, sensors (e.g., the sensor system **305**) in the battery **705** may measure voltage, temperature, and gas levels of the battery and the vehicle computing system **200** (e.g., the battery management system **360**) may use this data about the battery **705** (e.g., the battery data **365**) to predict that a thermal event **605** will occur. For example, the vehicle **105** may be parked and in an off state **620** in the user's garage. The temperature of the battery **705** of the vehicle **105** may be monitored by sensors of the vehicle **105** and the vehicle computing system **200** may detect that the temperature of the battery **705** is reaching a threshold level, such as at the upper end of the normal operating temperature range for the battery **705**, where the thermal event **605** is predicted to occur.

(97) The method **900** in an embodiment may include an operation **920**, in which the vehicle computing system **200** may output an initial message **610** indicating that the thermal event **605** is predicted to occur. The initial message **610** with the thermal event **605** prediction can be output to the computing devices **615** remote from the vehicle **105**. For instance, the vehicle computing system **200** may detect that the temperature of the battery **705** of the vehicle **105** that is parked and in the off state **620** in the garage of the user **120** is reaching a threshold level where a thermal event **605** is predicted to occur. Additionally, or alternatively, the vehicle computing system **200** may determine that the rate of change in the battery temperature is at a certain level (or is increasing) such that it appears the battery is approaching a thermal event. For example, the vehicle computing system **200** may determine that the temperature of the battery **705** increased by 25% in under five seconds and that a thermal event **605** is predicted to occur due to the temperature increase. The vehicle computing system **200** may output an initial message **610** to a computing device belonging to the user **120**, such as a phone, and the message **610** may indicate that the thermal event **605** is predicted to occur in the vehicle **105**.

(98) The method **900** in an embodiment may include an operation **925**, in which the vehicle computing system **200** may determine, based on the battery data **365**, an occurrence of a thermal event **605** onboard the vehicle **105** while the vehicle **105** is in the off state **620**. The thermal event **605** may be a thermal runaway **710** event.

(99) By way of example, sensors in the battery **705** of the vehicle **105** that is parked and in the off

state **620** in the garage of the user **120** may measure voltage, temperature, and gas levels of the battery **705**, which may be stored as battery data **365**. Sensors in the battery cells of the battery **705** of the vehicle **105** may measure the temperature of a battery cell and the gas levels of the battery cell, as well as the temperature and gas levels of the battery cells adjacent to the battery cell. The vehicle computing system **200** (e.g., the battery management system **360**) may access the battery data **365** that includes the temperature and gas levels of the battery cells and determine that the temperature and gas levels of the battery cell and the adjacent battery cells are at a level where thermal runaway **710** has occurred in the battery **705** onboard the vehicle **105**.

(100) The method **900** in an embodiment may include an operation **930**, in which the vehicle computing system **200**, in response to determining the occurrence of the thermal event **605**, generates a message **610** associated with the battery **705**. The message **610** may be indicative of the thermal event **635**, the state **640** of the battery **705**, or the location **645** of the vehicle **105**. For instance, the vehicle **105** may be parked and in an off state **620** in the user's garage and the vehicle computing device **200** may determine that a thermal event **605** occurred onboard the vehicle **105** based on the battery data. The thermal event **605** may be thermal runaway **710** due to an overheating of the battery **705** of the vehicle **105**. The vehicle computing device **200** can generate a message **610** that indicates that the state **640** of the battery is that the battery is overheating, that the thermal event **605** is thermal runaway **710**, and that the location **645** of the vehicle **105** is at the user's home address.

(101) The method **900** in an embodiment may include an operation **935**, in which the vehicle computing system **200** outputs, to one or more computing devices **615** remote from the vehicle **105**, the message **610** associated with the battery **705**. The computing devices **615** remote from the vehicle **105** may include one or more user devices **115** associated with one or more users **120** of the vehicle **105**, computing devices associated with a vehicle assistance service **650** or an emergency provider **655**, or computing devices included in a computing system onboard another vehicle **660**. For example, the vehicle computing device **200** may generate the message **610** that the state **640** of the battery is that the battery is overheating, that the thermal event **605** is thermal runaway **710**, and that the location **645** of the vehicle **105** is at the user's home address where the vehicle **105** is parked in the garage, and the message **610** with this information can be output to a user device **115**, such as a phone, of the user **120** and to an emergency provider **655**. In some examples, where the vehicle **105** is being transported by another vehicle **805**, the computing devices **615** remote from the vehicle **105** may include a computing device onboard the other vehicle **805** or a user device **115** of an operator of the other vehicle **805**.

(102) For example, the message **610** may be output to the user device **115** and the message **610** can include a hyperlink that will cause the user device **115** to call an emergency service **655**. Additionally, or alternatively, a user **120** can review the message **610** output to the user device **115** and contact an emergency provider **655** by telephone or text. In another example, the cloud platform can process the message **610** and can transmit a request for emergency services to another system (e.g., of an emergency provider **655**). Additionally, or alternatively, the cloud platform can wake up the vehicle **105**, transmit the message **610** to the vehicle **105** for display on the display device **345** to notify passengers inside the vehicle **105** of the thermal event **605**, and to take mitigation actions to stop the thermal event **605**. In another example, the message **610** may be processed by an emergency service provider **655**, which can dispatch emergency personnel in response thereto to the location **645** identified in the message **610**. In another example, the message **610** may be processed by a vehicle assistance service **650** that can send personnel to assist with the vehicle **105**. In some examples, the message **610** may be output to a computing system onboard another vehicle **660** and the message **610** can include information that will allow the passengers of the other vehicle **660** to move away from the vehicle **105** experiencing the thermal event **605**.

(103) In some examples, in operation **940**, the vehicle computing system **200** may output a signal to transition the vehicle **105** to an on state **625** or an awake state. The on state **625** of the vehicle

105 can provide power to components of the vehicle, such as the head unit **347** and the display device **345** of the vehicle **105**. In order to be able to display the message **610** on the display device **345** of the vehicle, the vehicle computing system **200** may output a signal to transition the vehicle **105** to the on state **625** and display the message **610** on the display device **345** to notify passengers to exit the vehicle **105**. Transitioning the vehicle **105** to the on state **625** can provide power that can allow the vehicle computing system **200** to take mitigation actions to attempt to stop the thermal event **605**. For instance, the thermal event **605** may be thermal runaway **710** due to overheating of the battery **705** and the vehicle **105** may transition to the on state **620** in order for the vehicle computing system **200** to attempt to cool off the battery **705** to prevent the spread of the thermal runaway **710** to other cells of the battery **705**.

(104) FIG. **10** illustrates a block diagram of an example computing system **5000** according to an embodiment hereof. The system **5000** includes a computing system **6005** (e.g., a computing system onboard a vehicle), a server computing system **7005** (e.g., a remote computing system, cloud computing platform), and a user device **8005** that are communicatively coupled over one or more networks **9050**.

(105) The computing system **6005** may include one or more computing devices **6010** or circuitry. For instance, the computing system **6005** may include a control circuit **6015** and a non-transitory computer-readable medium **6020**, also referred to herein as memory. In an embodiment, the control circuit **6015** may include one or more processors (e.g., microprocessors), one or more processing cores, a programmable logic circuit (PLC) or a programmable logic/gate array (PLA/PGA), a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), or any other control circuit. In some implementations, the control circuit **6015** may be part of, or may form, a vehicle control unit (also referred to as a vehicle controller) that is embedded or otherwise disposed in a vehicle (e.g., a Mercedes-Benz® car or van). For example, the vehicle controller may be or may include an infotainment system controller (e.g., an infotainment head-unit), a telematics control unit (TCU), an electronic control unit (ECU), a central powertrain controller (CPC), a charging controller, a central exterior & interior controller (CEIC), a zone controller, or any other controller. In an embodiment, the control circuit **6015** may be programmed by one or more computer-readable or computer-executable instructions stored on the non-transitory computer-readable medium **6020**.

(106) In an embodiment, the non-transitory computer-readable medium **6020** may be a memory device, also referred to as a data storage device, which may include an electronic storage device, a magnetic storage device, an optical storage device, an electromagnetic storage device, a semiconductor storage device, or any suitable combination thereof. The non-transitory computer-readable medium **6020** may form, e.g., a hard disk drive (HDD), a solid state drive (SDD) or solid state integrated memory, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a static random access memory (SRAM), dynamic random access memory (DRAM), a portable compact disc read-only memory (CD-ROM), a digital versatile disk (DVD), and/or a memory stick.

(107) The non-transitory computer-readable medium **6020** may store information that may be accessed by the control circuit **6015**. For instance, the non-transitory computer-readable medium **6020** (e.g., memory devices) may store data **6025** that may be obtained, received, accessed, written, manipulated, created, and/or stored. The data **6025** may include, for instance, any of the data or information described herein. In some implementations, the computing system **6005** may obtain data from one or more memories that are remote from the computing system **6005**.

(108) The non-transitory computer-readable medium **6020** may also store computer-readable instructions **6030** that may be executed by the control circuit **6015**. The instructions **6030** may be software written in any suitable programming language or may be implemented in hardware. The instructions may include computer-readable instructions, computer-executable instructions, etc. As described herein, in various embodiments, the terms “computer-readable instructions” and

"computer-executable instructions" are used to describe software instructions or computer code configured to carry out various tasks and operations. In various embodiments, if the computer-readable or computer-executable instructions form modules, the term "module" refers broadly to a collection of software instructions or code configured to cause the control circuit **6015** to perform one or more functional tasks. The modules and computer-readable/executable instructions may be described as performing various operations or tasks when the control circuit **6015** or other hardware component is executing the modules or computer-readable instructions.
(109) The instructions **6030** may be executed in logically and/or virtually separate threads on the control circuit **6015**. For example, the non-transitory computer-readable medium **6020** may store instructions **6030** that when executed by the control circuit **6015** cause the control circuit **6015** to perform any of the operations, methods and/or processes described herein. In some cases, the non-transitory computer-readable medium **6020** may store computer-executable instructions or computer-readable instructions, such as instructions to perform at least a portion of the method of FIG. **9**.
(110) The computing system **6005** may include one or more communication interfaces **6035**. The communication interfaces **6035** may be used to communicate with one or more other systems. The communication interfaces **6035** may include any circuits, components, software, etc. for communicating via one or more networks (e.g., networks **9050**). In some implementations, the communication interfaces **6035** may include for example, one or more of a communications controller, receiver, transceiver, transmitter, port, conductors, software and/or hardware for communicating data/information.
(111) The computing system **6005** may also include one or more user input components **6040** that receives user input. For example, the user input component **6040** may be a touch-sensitive component (e.g., a touch-sensitive display screen or a touch pad) that is sensitive to the touch of a user input object (e.g., a finger or a stylus). The touch-sensitive component may serve to implement a virtual keyboard. Other example user input components include a microphone, a traditional keyboard, cursor-device, joystick, or other devices by which a user may provide user input.
(112) The computing system **6005** may include one or more output components **6045**. The output components **6045** may include hardware and/or software for audibly or visually producing content. For instance, the output components **6045** may include one or more speakers, earpieces, headsets, handsets, etc. The output components **6045** may include a display device, which may include hardware for displaying a user interface and/or messages for a user. By way of example, the output component **6045** may include a display screen, CRT, LCD, plasma screen, touch screen, TV, projector, tablet, and/or other suitable display components.
(113) The server computing system **7005** may include one or more computing devices **7010**. In an embodiment, the server computing system **7005** may include or is otherwise implemented by one or more server computing devices. In instances in which the server computing system **7005** includes plural server computing devices, such server computing devices may operate according to sequential computing architectures, parallel computing architectures, or some combination thereof.
(114) The server computing system **7005** may include a control circuit **7015** and a non-transitory computer-readable medium **7020**, also referred to herein as memory **7020**. In an embodiment, the control circuit **7015** may include one or more processors (e.g., microprocessors), one or more processing cores, a programmable logic circuit (PLC) or a programmable logic/gate array (PLA/PGA), a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), or any other control circuit. In an embodiment, the control circuit **7015** may be programmed by one or more computer-readable or computer-executable instructions stored on the non-transitory computer-readable medium **7020**.
(115) In an embodiment, the non-transitory computer-readable medium **7020** may be a memory device, also referred to as a data storage device, which may include an electronic storage device, a magnetic storage device, an optical storage device, an electromagnetic storage device, a

semiconductor storage device, or any suitable combination thereof. The non-transitory computer-readable medium may form, e.g., a hard disk drive (HDD), a solid state drive (SDD) or solid state integrated memory, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a static random access memory (SRAM), dynamic random access memory (DRAM), a portable compact disc read-only memory (CD-ROM), a digital versatile disk (DVD), and/or a memory stick.

(116) The non-transitory computer-readable medium **7020** may store information that may be accessed by the control circuit **7015**. For instance, the non-transitory computer-readable medium **7020** (e.g., memory devices) may store data **7025** that may be obtained, received, accessed, written, manipulated, created, and/or stored. The data **7025** may include, for instance, any of the data or information described herein. In some implementations, the server computing system **7005** may obtain data from one or more memories that are remote from the server computing system **7005**.

(117) The non-transitory computer-readable medium **7020** may also store computer-readable instructions **7030** that may be executed by the control circuit **7015**. The instructions **7030** may be software written in any suitable programming language or may be implemented in hardware. The instructions may include computer-readable instructions, computer-executable instructions, etc. As described herein, in various embodiments, the terms “computer-readable instructions” and “computer-executable instructions” are used to describe software instructions or computer code configured to carry out various tasks and operations. In various embodiments, if the computer-readable or computer-executable instructions form modules, the term “module” refers broadly to a collection of software instructions or code configured to cause the control circuit **7015** to perform one or more functional tasks. The modules and computer-readable/executable instructions may be described as performing various operations or tasks when the control circuit **7015** or other hardware component is executing the modules or computer-readable instructions.

(118) The instructions **7030** may be executed in logically and/or virtually separate threads on the control circuit **7015**. For example, the non-transitory computer-readable medium **7020** may store instructions **7030** that when executed by the control circuit **7015** cause the control circuit **7015** to perform any of the operations, methods and/or processes described herein. In some cases, the non-transitory computer-readable medium **7020** may store computer-executable instructions or computer-readable instructions, such as instructions to perform at least a portion of the method of FIG. **9**.

(119) The server computing system **7005** may include one or more communication interfaces **7035**. The communication interfaces **7035** may be used to communicate with one or more other systems. The communication interfaces **7035** may include any circuits, components, software, etc. for communicating via one or more networks (e.g., networks **9050**). In some implementations, the communication interfaces **7035** may include for example, one or more of a communications controller, receiver, transceiver, transmitter, port, conductors, software and/or hardware for communicating data/information.

(120) The computing system **6005** and/or the server computing system **7005** may also be in communication with a user device **8005** that is communicatively coupled over the networks **9050**.

(121) The user device **8005** may include one or more computing devices **8010**. The user device **8005** may include a control circuit **8015** and a non-transitory computer-readable medium **8020**, also referred to herein as memory **8020**. In an embodiment, the control circuit **8015** may include one or more processors (e.g., microprocessors), one or more processing cores, a programmable logic circuit (PLC) or a programmable logic/gate array (PLA/PGA), a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), or any other control circuit. In an embodiment, the control circuit **8015** may be programmed by one or more computer-readable or computer-executable instructions stored on the non-transitory computer-readable medium **8020**.

(122) In an embodiment, the non-transitory computer-readable medium **8020** may be a memory device, also referred to as a data storage device, which may include an electronic storage device, a

magnetic storage device, an optical storage device, an electromagnetic storage device, a semiconductor storage device, or any suitable combination thereof. The non-transitory computer-readable medium may form, e.g., a hard disk drive (HDD), a solid state drive (SDD) or solid state integrated memory, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a static random access memory (SRAM), dynamic random access memory (DRAM), a portable compact disc read-only memory (CD-ROM), a digital versatile disk (DVD), and/or a memory stick.

(123) The non-transitory computer-readable medium **8020** may store information that may be accessed by the control circuit **8015**. For instance, the non-transitory computer-readable medium **8020** (e.g., memory devices) may store data **8025** that may be obtained, received, accessed, written, manipulated, created, and/or stored. The data **8025** may include, for instance, any of the data or information described herein. In some implementations, the user device **8005** may obtain data from one or more memories that are remote from the user device **8005**.

(124) The non-transitory computer-readable medium **8020** may also store computer-readable instructions **8030** that may be executed by the control circuit **8015**. The instructions **8030** may be software written in any suitable programming language or may be implemented in hardware. The instructions may include computer-readable instructions, computer-executable instructions, etc. As described herein, in various embodiments, the terms “computer-readable instructions” and “computer-executable instructions” are used to describe software instructions or computer code configured to carry out various tasks and operations. In various embodiments, if the computer-readable or computer-executable instructions form modules, the term “module” refers broadly to a collection of software instructions or code configured to cause the control circuit **8015** to perform one or more functional tasks. The modules and computer-readable/executable instructions may be described as performing various operations or tasks when the control circuit **8015** or other hardware component is executing the modules or computer-readable instructions.

(125) The instructions **8030** may be executed in logically or virtually separate threads on the control circuit **8015**. For example, the non-transitory computer-readable medium **8020** may store instructions **8030** that when executed by the control circuit **8015** cause the control circuit **8015** to perform any of the operations, methods and/or processes described herein. In some cases, the non-transitory computer-readable medium **8020** may store computer-executable instructions or computer-readable instructions, such as instructions to perform at least a portion of the method of FIG. **9**.

(126) The user device **8005** may include one or more communication interfaces **8035**. The communication interfaces **8035** may be used to communicate with one or more other systems. The communication interfaces **8035** may include any circuits, components, software, etc. for communicating via one or more networks (e.g., networks **9050**). In some implementations, the communication interfaces **8035** may include for example, one or more of a communications controller, receiver, transceiver, transmitter, port, conductors, software and/or hardware for communicating data/information.

(127) The user device **8005** may also include one or more user input components **8040** that receives user input. For example, the user input component **8040** may be a touch-sensitive component (e.g., a touch-sensitive display screen or a touch pad) that is sensitive to the touch of a user input object (e.g., a finger or a stylus). The touch-sensitive component may serve to implement a virtual keyboard. Other example user input components include a microphone, a traditional keyboard, cursor-device, joystick, or other devices by which a user may provide user input.

(128) The user device **8005** may include one or more output components **8045**. The output components **8045** may include hardware and/or software for audibly or visually producing content. For instance, the output components **8045** may include one or more speakers, earpieces, headsets, handsets, etc. The output components **8045** may include a display device, which may include hardware for displaying a user interface and/or messages for a user. By way of example, the output

component **8045** may include a display screen, CRT, LCD, plasma screen, touch screen, TV, projector, tablet, and/or other suitable display components.

(129) The one or more networks **9050** may be any type of communications network, such as a local area network (e.g., intranet), wide area network (e.g., Internet), or some combination thereof and may include any number of wired or wireless links. In general, communication over a network **9050** may be carried via any type of wired and/or wireless connection, using a wide variety of communication protocols (e.g., TCP/IP. HTTP, SMTP, FTP), encodings or formats (e.g., HTML, XML), and/or protection schemes (e.g., VPN, secure HTTP, SSL).

ADDITIONAL DISCUSSION OF VARIOUS EMBODIMENTS

(130) Embodiment 1 relates to a computing system of a vehicle. In this embodiment, the computing system includes a control circuit configured to obtain, while the vehicle is in an off state, battery data indicative of a state of a battery onboard the vehicle. The control circuit is configured to determine, based on the battery data, an occurrence of a thermal event onboard the vehicle while the vehicle is in the off state. The control circuit is configured to in response to determining the occurrence of the thermal event, generate a message associated with the battery. The control circuit is configured to output, to one or more computing devices remote from the vehicle, the message associated with the battery.

(131) Embodiment 2 includes the computing system of Embodiment 1. In this embodiment,

(132) the thermal event is a thermal runaway event.

(133) Embodiment 3 includes the computing system of any of embodiments 1 or 2. In this embodiment, the control circuit is further configured to determine that the vehicle has transitioned from an on state to the off state.

(134) Embodiment 4 includes the computing system of any of embodiments 1 to 3. In this embodiment, the vehicle is in a first battery monitoring mode when the vehicle is in the on state and a second battery monitoring mode when the vehicle is in the off state. The second battery monitoring mode is different from the first battery monitoring mode. The control circuit is configured to output the message associated with the battery to the one or more computing devices remote from the vehicle when the vehicle is in the second battery monitoring mode.

(135) Embodiment 5 includes the computing system of any of embodiments 1 to 4. In this embodiment, the vehicle is in a parked state.

(136) Embodiment 6 includes the computing system of any of embodiments 1 to 5. In this embodiment, the message is indicative of the thermal event.

(137) Embodiment 7 includes the computing system of any of embodiments 1 to 6. In this embodiment, the message is indicative of the state of the battery.

(138) Embodiment 8 includes the computing system of any of embodiments 1 to 7. In this embodiment, the one or more computing devices remote from the vehicle are a plurality of user devices associated with users of the vehicle.

(139) Embodiment 9 includes the computing system of any of embodiments 1 to 8. In this embodiment, the one or more computing devices remote from the vehicle are associated with a vehicle assistance service or an emergency provider.

(140) Embodiment 10 includes the computing system of any of embodiments 1 to 9. In this embodiment, the one or more computing devices remote from the vehicle are included in a computing system onboard another vehicle.

(141) Embodiment 11 relates to a computer-implemented method. In this embodiment, the computer-implemented method includes obtaining, while a vehicle is in an off state, battery data indicative of a state of a battery onboard the vehicle. The computer-implemented method includes determining, based on the battery data, an occurrence of a thermal event onboard the vehicle while the vehicle is in the off state. The computer-implemented method includes in response to determining the occurrence of the thermal event, generating a message associated with the battery. The computer-implemented method includes outputting, to one or more computing devices remote

from the vehicle, the message associated with the battery.
(142) Embodiment 12 includes the computer-implemented method of Embodiment 11. In this embodiment, the computer-implemented method includes determining that the vehicle has transitioned from an on state to the off state.
(143) Embodiment 13 includes the computer-implemented method of any of Embodiment 11 or 12. In this embodiment, the computer-implemented method includes predicting that the thermal event will occur and outputting an initial message indicating that the thermal event is predicted to occur to the one or more computing devices remote from the vehicle.
(144) Embodiment 14 includes the computer-implemented method of any of Embodiments 11 to 13. In this embodiment, the computer-implemented method includes outputting a signal to transition the vehicle to an on state or an awake state.
(145) Embodiment 15 includes the computer-implemented method of any of Embodiments 11 to 14. In this embodiment, the one or more computing devices remote from the vehicle include at least one of: (i) a user device, (ii) a computing device associated with a vehicle assistance service, or (iii) a computing device associated with an emergency provider.
(146) Embodiment 16 includes the computer-implemented method of any of Embodiments 11 to 15. In this embodiment, the one or more computing devices remote from the vehicle are a plurality of user devices associated with users of the vehicle.
(147) Embodiment 17 includes the computer-implemented method of any of Embodiments 11 to 16. In this embodiment, the vehicle is being transported by another vehicle. The one or more computing devices remote from the vehicle are a computing device onboard the other vehicle or a user device of an operator of the other vehicle.
(148) Embodiment 18 includes the computer-implemented method of any of Embodiments 11 to 17. In this embodiment, the message is indicative of at least one of: (i) the thermal event, or (ii) the state of the battery.
(149) Embodiment 19 includes the computer-implemented method of any of Embodiments 11 to 18. In this embodiment, the message is indicative of a location of the vehicle.
(150) Embodiment 20 relates to one or more non-transitory computer-readable media that store instructions that are executable by a control circuit to: obtain, while a vehicle is in an off state, battery data indicative of a state of a battery onboard the vehicle; determine, based on the battery data, an occurrence of a thermal event onboard the vehicle while the vehicle is in the off state; in response to determining the occurrence of the thermal event, generate a message associated with the battery; and output, to one or more computing devices remote from the vehicle, the message associated with the battery.
ADDITIONAL DISCLOSURE
(151) As used herein, adjectives and their possessive forms are intended to be used interchangeably unless apparent otherwise from the context and/or expressly indicated. For instance, “component of a/the vehicle” may be used interchangeably with “vehicle component” where appropriate. Similarly, words, phrases, and other disclosure herein is intended to cover obvious variants and synonyms even if such variants and synonyms are not explicitly listed.
(152) The technology discussed herein makes reference to servers, databases, software applications, and other computer-based systems, as well as actions taken and information sent to and from such systems. The inherent flexibility of computer-based systems allows for a great variety of possible configurations, combinations, and divisions of tasks and functionality between and among components. For instance, processes discussed herein may be implemented using a single device or component or multiple devices or components working in combination. Databases and applications may be implemented on a single system or distributed across multiple systems. Distributed components may operate sequentially or in parallel.
(153) While the present subject matter has been described in detail with respect to various specific example embodiments thereof, each example is provided by way of explanation, not limitation of

the disclosure. Those skilled in the art, upon attaining an understanding of the foregoing, may readily produce alterations to, variations of, and equivalents to such embodiments. Accordingly, the subject disclosure does not preclude inclusion of such modifications, variations and/or additions to the present subject matter as would be readily apparent to one of ordinary skill in the art. For instance, features illustrated or described as part of one embodiment may be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present disclosure cover such alterations, variations, and equivalents.
(154) Aspects of the disclosure have been described in terms of illustrative implementations thereof. Numerous other implementations, modifications, or variations within the scope and spirit of the appended claims may occur to persons of ordinary skill in the art from a review of this disclosure. Any and all features in the following claims may be combined or rearranged in any way possible. Accordingly, the scope of the present disclosure is by way of example rather than by way of limitation, and the subject disclosure does not preclude inclusion of such modifications, variations or additions to the present subject matter as would be readily apparent to one of ordinary skill in the art. Moreover, terms are described herein using lists of example elements joined by conjunctions such as "and," "or," "but," etc. It should be understood that such conjunctions are provided for explanatory purposes only. The term "or" and "and/or" may be used interchangeably herein. Lists joined by a particular conjunction such as "or," for example, may refer to "at least one of" or "any combination of" example elements listed therein, with "or" being understood as "and/or" unless otherwise indicated. Also, terms such as "based on" should be understood as "based at least in part on."
(155) Those of ordinary skill in the art, using the disclosures provided herein, will understand that the elements of any of the claims, operations, or processes discussed herein may be adapted, rearranged, expanded, omitted, combined, or modified in various ways without deviating from the scope of the present disclosure. At times, elements may be listed in the specification or claims using a letter reference for exemplary illustrated purposes and is not meant to be limiting. Letter references, if used, do not imply a particular order of operations or a particular importance of the listed elements. For instance, letter identifiers such as (a), (b), (c), . . . , (i), (ii), (iii), . . . , etc. may be used to illustrate operations or different elements in a list. Such identifiers are provided for the case of the reader and do not denote a particular order, importance, or priority of steps, operations, or elements. For instance, an operation illustrated by a list identifier of (a), (i), etc. may be performed before, after, or in parallel with another operation illustrated by a list identifier of (b), (ii), etc.

Claims

1. A computing system of a vehicle comprising: a control circuit configured to: obtain, while the vehicle is in an off state, battery data indicative of a state of a battery onboard the vehicle; determine, based on the battery data, an occurrence of a thermal event onboard the vehicle while the vehicle is in the off state, wherein the thermal event is determined by: monitoring one or more threshold values associated with an on state for the vehicle and one or more threshold values associated with the off state for the vehicle, wherein the threshold values are different when the vehicle is in the on state and the off state; and predicting the thermal event by implementing the appropriate threshold values; in response to determining the occurrence of the thermal event, generate a message associated with the battery; and output, to one or more computing devices remote from the vehicle, the message associated with the battery.
2. The computing system of claim 1, wherein the thermal event comprises a thermal runaway event.
3. The computing system of claim 1, wherein the control circuit is further configured to: determine that the vehicle has transitioned from an on state to the off state.

4. The computing system of claim 3, wherein the vehicle is in a first battery monitoring mode when the vehicle is in the on state and a second battery monitoring mode when the vehicle is in the off state, the second battery monitoring mode being different from the first battery monitoring mode, and wherein the control circuit is configured to output the message associated with the battery to the one or more computing devices remote from the vehicle when the vehicle is in the second battery monitoring mode.
5. The computing system of claim 1, wherein the vehicle is in a parked state.
6. The computing system of claim 1, wherein the message is indicative of the thermal event.
7. The computing system of claim 1, wherein the message is indicative of the state of the battery.
8. The computing system of claim 1, wherein the one or more computing devices remote from the vehicle comprises a plurality of user devices associated with users of the vehicle.
9. The computing system of claim 1, wherein the one or more computing devices remote from the vehicle are associated with a vehicle assistance service or an emergency provider.
10. The computing system of claim 1, wherein the one or more computing devices remote from the vehicle are included in a computing system onboard another vehicle.
11. A computer-implemented method comprising: obtaining, while a vehicle is in an off state, battery data indicative of a state of a battery onboard the vehicle, wherein the vehicle is being transported by another vehicle; determining, based on the battery data, an occurrence of a thermal event onboard the vehicle while the vehicle is in the off state; in response to determining the occurrence of the thermal event, generating a message associated with the battery; and outputting, to one or more computing devices remote from the vehicle, the message associated with the battery, wherein the one or more computing devices remote from the vehicle comprise a computing device onboard the other vehicle or a user device of an operator of the other vehicle.
12. The computer-implemented method of claim 11, further comprising: determining that the vehicle has transitioned from an on state to the off state.
13. The computer-implemented method of claim 11, further comprising: predicting that the thermal event will occur; and outputting, to the one or more computing devices remote from the vehicle, an initial message indicating that the thermal event is predicted to occur.
14. The computer-implemented method of claim 11, further comprising: outputting a signal to transition the vehicle to an on state or an awake state.
15. The computer-implemented method of claim 11, wherein the one or more computing devices remote from the vehicle comprise at least one of: (i) a user device, (ii) a computing device associated with a vehicle assistance service, or (iii) a computing device associated with an emergency provider.
16. The computer-implemented method of claim 11, wherein the one or more computing devices remote from the vehicle comprise a plurality of user devices associated with users of the vehicle.
17. The computer-implemented method of claim 11, wherein the message is indicative of at least one of: (i) the thermal event, or (ii) the state of the battery.
18. The computer-implemented method of claim 11, wherein the message is indicative of a location of the vehicle.
19. One or more non-transitory computer-readable media that store instructions that are executable by a control circuit to: obtain, while a vehicle is in an off state, battery data indicative of a state of a battery onboard the vehicle; determine, based on the battery data, an occurrence of a thermal event onboard the vehicle while the vehicle is in the off state, wherein the thermal event is determined by: monitoring one or more threshold values associated with an onstate for the vehicle and one or more threshold values associated with the off state for the vehicle, wherein the threshold values are different when the vehicle is in the on state and the off state; and predicting the thermal event by implementing the appropriate threshold values; in response to determining the occurrence of the thermal event, generate a message associated with the battery; and output, to one or more computing devices remote from the vehicle, the message associated with the battery.

20. The computing system of claim 1, wherein the one or more thresholds is indicative of at least one of: i) operating temperature range for the battery, ii) gas levels of the battery cell, iii) pressure of the battery or battery cell, iv) charge level, v) voltage, vi) capacity, vii) leakage, viii) gas levels, ix) internal resistance, or x) state of the battery case.